



## Appendices

## Appendix: Pipeline Stage Reference I

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Table 1: Single-project DAG exported from default pipeline.yaml (names shown in topological order). Scripts live under scripts/.

| Stage name               | Script / method                                                                | Primary inputs                                                 | Outputs / artefacts             | Failure mode                                       |
|--------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------|----------------------------------------------------|
| Clean Output Directories | _run_clean_outputs                                                             | prior projects/<name>/output/, mirrored output/<name>/ targets | emptied trees                   | Blocking                                           |
| Environment Setup        | scripts/pipeline/stage_00_setup.py                                             | toolchain probes                                               | scaffold dirs, env exports      | Blocking                                           |
| Infrastructure Tests     | scripts/pipeline/stage_01_test.py --infr / a-only --infra-scope pipeline-smoke | tests/infra_tests                                              | coverage + junit-style logs     | tolerant ceilings                                  |
| Project Tests            | scripts/pipeline/stage_01_test.py --project-only                               | projects/<name>/tests/                                         | stage artefacts                 | blocking by default                                |
| Project Analysis         | scripts/pipeline/stage_02_analysis.py                                          | thin scripts                                                   | figures/, data/, reports        | Blocking                                           |
| Connector Search         | scripts/pipeline/stage_08_connector_search.py                                  | manuscript/config.yaml                                         | output/data/connector_search/   | opt-in (science tag); skipped if not configured    |
| Provenance Record        | scripts/pipeline/stage_09_provenance_record.py --stage "Connector Search"      | prior stage outputs                                            | .provenance/graph.json          | opt-in (provenance tag); skipped if not configured |
| PDF Rendering            | scripts/pipeline/stage_03_render.py                                            | manuscript/, placeholders                                      | .pdf/.tex bundles               | Blocking                                           |
| Output Validation        | scripts/pipeline/stage_04_validate.py                                          | render tree                                                    | Markdown + PDF diagnostics JSON | Blocking / downgrade                               |

## Appendix: Pipeline Stage Reference II

|                       |                                         |                               |                                  |                                  |
|-----------------------|-----------------------------------------|-------------------------------|----------------------------------|----------------------------------|
| LLM Scientific Review | scripts/pipeline/stage_06_llm_review.py | resolved manuscript artefacts | textual reviews                  | Optional skip (allow_skip)       |
| LLM Translations      | scripts/pipeline/stage_06_llm_review.py | abstract metadata             | multilingual snippets            | Optional skip (allow_skip)       |
| Copy Outputs          | scripts/pipeline/stage_05_copy.py       | validated tree                | mirrored output/<name>/...       | soft fail logged                 |
| Ebook Generation      | scripts/pipeline/stage_11_ebook.py      | rendered combined markdown    | output/ebook/ (EPUB/MOBI/DOCX)   | opt-in (ebook tag); soft fail    |
| Metadata              | scripts/pipeline/stage_12_metadata.py   | manuscript/config .yaml       | output/metadata/ (ONIX/JSON/OPF) | opt-in (metadata tag); soft fail |
| Package Executable    | scripts/runner/bundle_executable.py     | project tree + outputs        | container bundle manifest        | opt-in (bundle tag)              |
| Archival Publication  | scripts/runner/archival_publication.py  | bundle + deliverables         | archival deposit manifest        | opt-in (archival tag)            |

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## Appendix: Pipeline Stage Reference I

`scripts/pipeline/stage_07_executive_report.py` is invoked **outside** this DAG whenever `execute_multi_project.py` aggregates pipelines—supplying cross-project KPI dashboards absent from lone-project checkpoints.